

Integration

Evaluate each indefinite integral.

1) $\int \frac{1}{\sqrt{16-x^2}} dx$

$$\sin^{-1} \frac{x}{4} + C$$

2) $\int \frac{1}{4+x^2} dx$

$$\frac{1}{2} \cdot \tan^{-1} \frac{x}{2} + C$$

3) $\int \frac{1}{x\sqrt{x^2-1}} dx$

$$\sec^{-1} |x| + C$$

4) $\int \frac{1}{16+x^2} dx$

$$\frac{1}{4} \cdot \tan^{-1} \frac{x}{4} + C$$

5) $\int \frac{1}{x\sqrt{x^2-4}} dx$

$$\frac{1}{2} \cdot \sec^{-1} \frac{|x|}{2} + C$$

6) $\int \frac{1}{\sqrt{25-x^2}} dx$

$$\sin^{-1} \frac{x}{5} + C$$

7) $\int \frac{1}{x\sqrt{x^2-81}} dx$

$$\frac{1}{9} \cdot \sec^{-1} \frac{|x|}{9} + C$$

8) $\int \frac{1}{4+x^2} dx$

$$\frac{1}{2} \cdot \tan^{-1} \frac{x}{2} + C$$